



JavaScript Essentials Training Assignments

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RECORD OF CHANGES

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2					

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CODE	:	<JS.E_Assignment04_Op1>
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LOC	:	<N/A>
DURATION	:	<180 minutes>

Exercise 1: DOM

Objectives:

Able to interact with HTML document using DOM API

Project Structure:

1. Create a new folder called **JS.E_Assignment04** (this will be your top folder)
2. Create folder **ex1_task1** and put every file related to this problem inside
3. Create folder **ex1_task2** and put every file related to this problem inside
4. Create folder **ex1_task3** and put every file related to this problem inside

Technical Requirement

Task 1:

1. Your task is to use DOM API to query the following DOM element:
 - *div at line 13 (in HTML file) - save it to an variable called `imageContainer`*
 - *Back button at line 18 (in HTML file) – save it to an variable called `backBtn`*
 - *Next button at line 19 (in HTML file) – save it to an variable called `nextBtn`*

Task 2:

Create a function called `renderImage()` that set the innerHTML of `imageContainer` to an HTML img with src is the variable `images`

Task 3:

1. In body of function callback of `backBtn` you have to do the following:
 - *Decrease value of variable **index** by 1*
 - *Check if that variable is less then 0 then set it to 0*
 - *Call the function `renderImage()` from Task 2*
2. In body of function callback of `nextBtn` you have to do the following:
 - *Increase value of variable `index` by 1*
 - *Check if that variable is greater or equal the length of array `images` then set `index` to the length of `images` - 1*
 - *Call the function `renderImage()` from Task 2*

3. Expected output:

Image Slideshow

[Back](#)[Next](#)

Exercise 2: Functions

Objectives:

Understand function feature of JavaScript and able to create your own function that solve common problem

Project Structure

In folder **JS.E_Assignment04** (this will be your top folder), create

1. Folder **ex2_task1** and put every file related to this problem inside
2. Folder **ex2_task2** and put every file related to this problem inside
3. Folder **ex2_task 3** and put every file related to this problem inside,...

Technical Requirement

Task 1:

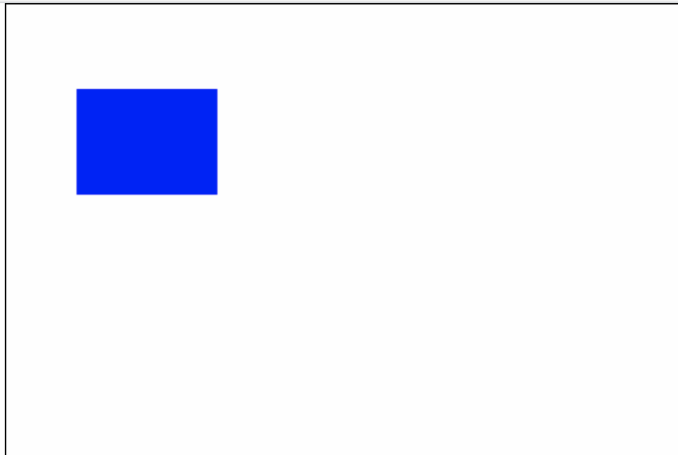
1. For the first task, you have to create a simple function — `chooseName()` — that prints a random name from the provided array (names) to the provided paragraph (para), and then run it once.
2. Expected output:

Anne

Task 2:

- 1 For our second functions-related task, you need to create a function that draws a rectangle on the provided `<canvas>` (reference variable `canvas`, context available in `ctx`), based on the five provided input variables:
 - *x* — the *x* coordinate of the rectangle.
 - *y* — the *y* coordinate of the rectangle.
 - *width* — the width of the rectangle.
 - *height* — the height of the rectangle.
 - *color* — the color of the rectangle.
2. You'll want to clear the canvas before drawing, so that when the code is updated in the case of the live version, you don't get lots of rectangles drawn on top of one another.

3. Expected output:

**Task 3:**

1. In this task, you return to the problem posed in Problem 1, with the aim of improving it.

The three improvements we want you to make are:

- Refactor the code that generates the random number into a separate function called `random()`, which takes as parameters two generic bounds that the random number should be between, and returns the result.
- Update the `chooseName()` function so that it makes use of the random number function, takes the array to choose from as a parameter (making it more flexible), and returns the result.
- Print the returned result into the paragraph (`para`)'s `textContent`.

2. Expected Output

Bert

Mark Scale

Functionality	50%
Code Quality	30%
Maintainable	20%